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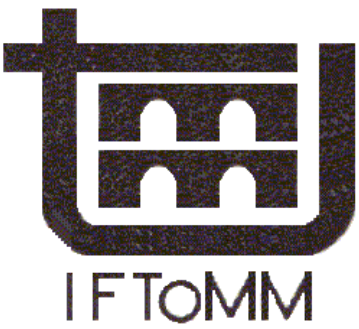
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PERFORMANCE COMPARISON OF ELECTROSTATIC FILM MOTORS WITH DIFFERENT SURFACE TREATMENTS TO REDUCE FRICTION

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ABSTRACT
In previous research of electrostatic film motors, glass beads were used to reduce the friction between films. However, it is difficult to keep the glass beads distributing evenly during long-term operation. Therefore, this work evaluates other two methods with potentiality to reduce the friction. In the first method, glass beads were coated on the film surface by using spin coating technology. In the second method, cover layers of the motor were replaced with PTFE films that have low friction. The experimental results verified that both methods can work well for friction reduction, although the durability for the latter method must be improved.

KEY WORDS: Electrostatic Film Motor, Glass Beads, Friction, Surface Treatment

1. INTRODUCTION

Since electrostatic motors are known to have higher performance than electromagnetic motors in micro-scale structures, they have been widely applied in micro-electromechanical system (MEMS) and micro-robotics [1, 2]. At the same time, macro-scale electrostatic film motors with powerful performance have also been developed, in which fine-patterned electrodes were embedded in films or thin boards with a size ranging from centimeters to tens of centimeters. Various operation principles for such electrostatic film motors have been proposed, which can be categorized into synchronous-type and induction-type [3-6]. The typical synchronous electrostatic film motor is called *dual excitation multiphase electrostatic drive* (DEMED) [5]. The motors are expected to be used as artificial muscle due to their unique features, such as mechanical flexibility and powerful performance. As the maximum output force of electrostatic film motor is proportional to the amplitude of the applied AC voltage, it is possible to provide excellent performance under high voltage operation. For such high voltage operation, the motor is operated in dielectric fluid to avoid discharge in the air.

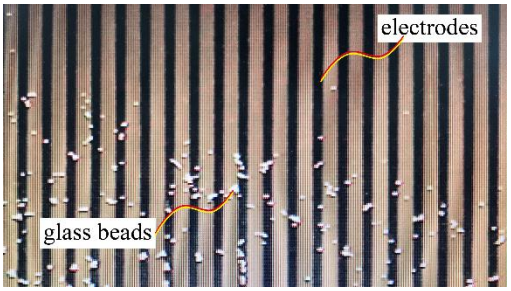


Fig. 1 Glass beads distribution after long-term operation

As the films rub against each other in the motor, previous studies mainly utilized glass beads to reduce the friction between the films. The beads are spread between the films and are supposed to form rolling friction between the glass beads and the films to reduce friction of the motor. However, it is difficult to keep the glass beads distribution, especially after long-term operation, as shown in Fig. 1. After operation, the distribution of glass beads can become uneven, with more beads in some places and less in other places. As a result, the overall friction becomes large, which could reduce the performance of the electrostatic film motor and can even make the motor unable to work. Therefore, this work evaluates two alternatives that can not only reduce friction, but also potentially realize longer-term stability. The details of the methods will be introduced in the next section. In Section 3, the performance of electrostatic film motors under different conditions were evaluated in experiments. Conclusions are given in the last section with some discussions.

2. ELECTROSTATIC FILM MOTORS

2・1 Principle of electrostatic film motors

The electrostatic film motor consists of a pair of films, the stator, and the slider. In the typical structure of DEMED, 3-phase parallel electrodes are embedded in plastic films, as shown in Fig. 2, which are typically realized by using flexible printed circuit technology. When 3-phase AC high voltage is applied to the motor, as in the figure, voltage potential waves are excited in both films and move towards different directions. Electrostatic force is excited between the two potential waves, whose horizontal component propels the slider in a synchronous manner. Theoretically, the thrust force is proportional to the square of the applied voltage amplitude.

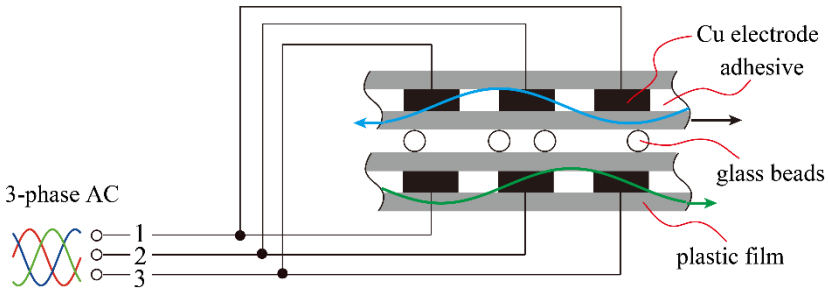


Fig. 2 Principle and structure of DEMED

2・2 Surface treatments of the motor

In this work, three different types of films were fabricated, as shown in Fig. 3. Those three types are all based on flexible printed circuit (FPC) film. As a base film, Liquid Crystal Polymer (LCP) was used [7, 8]. In type (a), which is a traditional type, both the base (bottom) layer and the cover (top) layer are LCP films, between which copper electrodes were embedded, as shown in Fig. 3 (a). In this type, glass beads with 20-μm diameter are sprinkled between the two films for operation.

The other methods to reduce friction are shown in Fig. 2 (b) and (c). In (b), 20-μm-diameter glass beads are evenly coated on the surface of the slider film by using spin coating with liquid tetrafluoroethylene (TFE), while there is no surface treatment on the stator film. As the glass beads are fixed on the surface, glass beads distribution could be kept for long time. In (c), polytetrafluoroethylene (PTFE) film (0.025 mm thick) that has low friction is used to replace the LCP cover layer (0.025 mm thick). PTFE film was glued onto a FPC film without a cover layer. As PTFE films alone can realize low friction, glass beads are not utilized for this type.

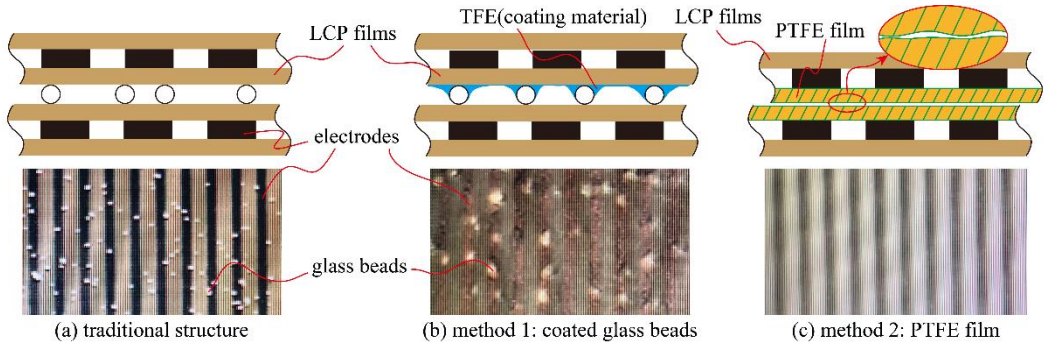


Fig. 3 Three surface treatments for friction reduction

3. EXPERIMENTAL EVALUATION

3・1 Experiment setup

Experiment setup for evaluating the thrust force performance is shown in Fig. 4. In all the three types, both stator and slider films share the same electrode structure with an electrode pitch of 0.2 mm and the electrode area of 50 mm × 150 mm. One end of the film was fixed on a plastic plate, while the other end was free to move. A digital signal processing system (D-space, ds1104) was used to generate 3-phase sinusoidal signal, which was amplified by a set of three high voltage amplifiers (NF, HVA4321) before being applied to the electrodes. The connection between the electrodes and the amplifiers is as shown in Fig. 2, and the motor was operated in dielectric fluid FC-770 (3M). A load cell (Kyowa, LVS-2KA) was used to measure the thrust force of the motors. The frequency of the applied AC volage was 5 Hz, and the voltage amplitude was changed from 500 to 1500 volts with an interval of 100 volts. Since the slider was fixed to the load cell, the application of sinusoidal voltage creates oscillating thrust force, whose peak value was recorded as the thrust force. However, as the peak value varies with the slider position due to intrinsic force ripple, the force was measured ten times within the range of one electrode pitch. The position shift was represented as *dx*, which was changed from 0 to 0.18 mm with an interval of 0.02 mm. The

effective area between the two films was 50 mm × (145+*dx*) mm, which was regarded as constant in the experiments, as *dx* is much smaller than 145 mm. The thrust forces of the motors with different surface treatments in Fig. 3 were measured respectively.

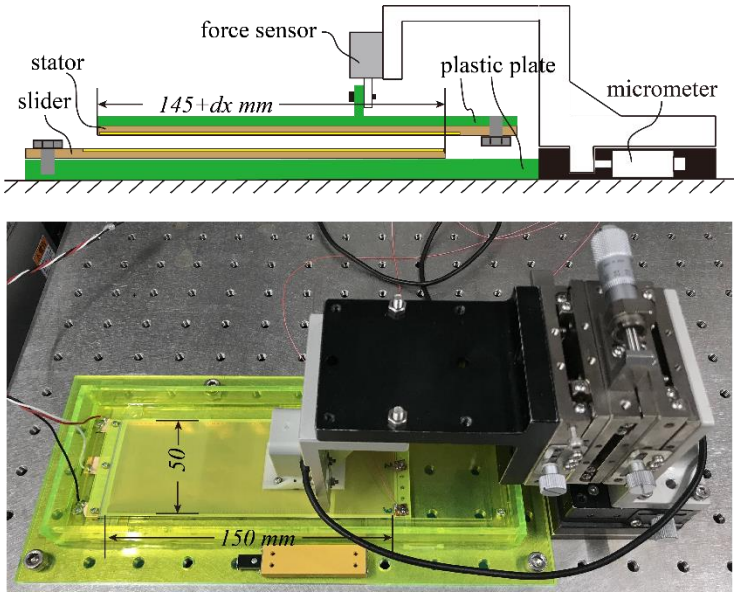


Fig. 4 Experiment setup

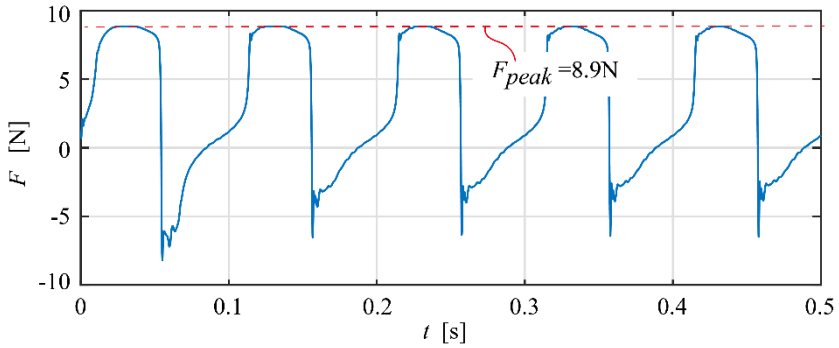


Fig. 5 Measured thrust force when the amplitude of the applied voltage is $V_{0-p} = 1.5$ kV

3・2 Results

The experimental results are shown in Fig. 5 and Fig. 6. When the voltage amplitude is $V_{0-p} = 1.5$ kV, the thrust force of the motor with the traditional structure was obtained as shown in Fig. 5. The peak value of the thrust force was described as F_{peak}. When the slider position was shifted, the value of F_{peak} varied, as shown in Fig. 6 using black markers. The peak values of thrust force under different voltage amplitudes were in the range between two black lines, as shown in Fig. 6. The thrust force of the motor with "coated glass beads" was also measured under different voltage amplitudes and different slider positions, which lie between two blue lines in Fig. 6. For the experiment using PTFE films, the thrust force under different voltages was only measured at one

position, because the film was broken during the measurement. The values are shown in Fig. 6 using red triangles.

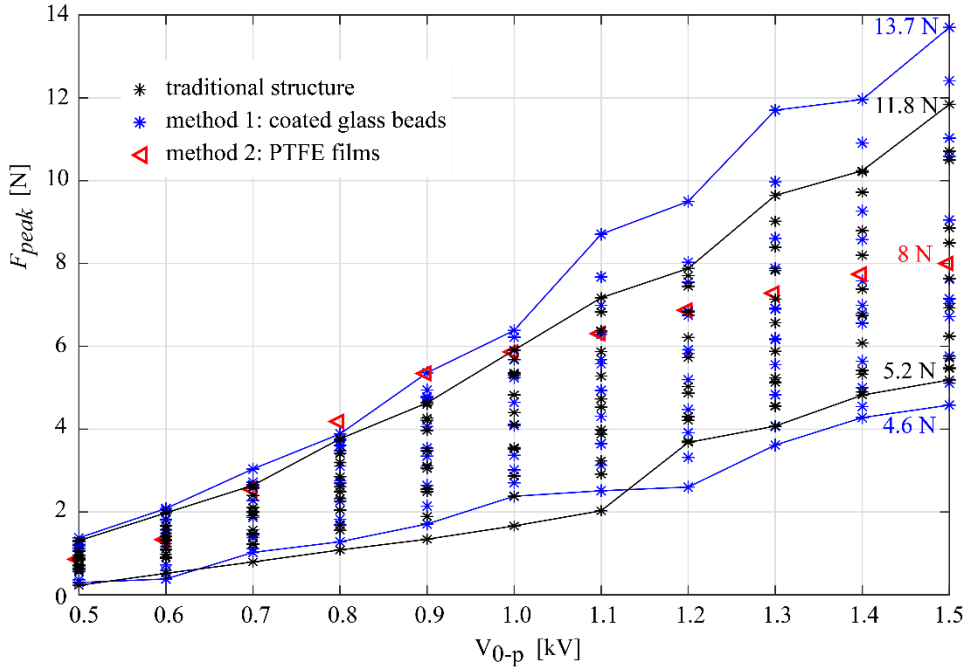


Fig. 6 Measured peak values of the thrust force under different conditions

4. DISCUSSION AND CONCLUSION

From Fig. 6, it was observed that the thrust force distribution of the motor with "coated glass beads" basically coincides with that of the motor with traditional structure. It means that the coated glass beads can perform almost the same as the sprinkled glass beads in the traditional structure, as far as static force is concerned. The PTFE film motor had poor lifetime and deteriorated quite easily, as shown in Fig. 7. When the PTFE film was immersed in the dielectric fluid, the PTFE film began to separate from the substrate, making wrinkles on the surface. As a result, the total gap between the films became too large for the motor to work well. However, from the measured thrust force of the PTFE film motor, it was confirmed that the peak value of the thrust force is in the same range as other motors, which means PTFE films would have a comparable friction reduction performance.

The experimental results confirmed that the expected methods could work well. Although the evaluations are still preliminary, the results show the possibility to replace the traditional structure with the expected methods. Considering the simplicity of the PTFE film motor, it would have significant potential to improve the performance of the electrostatic film motor. This time, the adhesion between the PTFE film and the substrate was not strong enough. In the future, the processing of PTFE film needs to be focused on to improve the durability. As the experiments were done quasi-statically, the performance in dynamic situations, where slider runs at practical speeds, should also be evaluated in the future study.

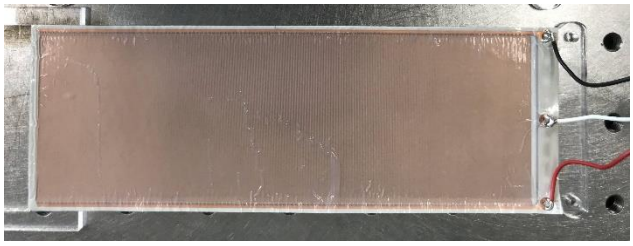


Fig. 7 PTFE film immersed in dielectric fluid for a long time. Wrinkles were created on its surface.

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